

Kochen–Specker uncolorability of the cyclotomic ray family $S_n \subset \mathbb{C}^3$ occurs if and only if $6 \mid n$

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We prove that the ray set S_n generated by the n th roots of unity in $\mathbb{C}^3$ is Kochen–Specker uncolorable if and only if $6 \mid n$. Sufficiency follows from the embedding into S_6 of an explicit 33-ray KS set with coordinates in $\{0\} \cup \mu_6$. Necessity is proved by analysis of short vanishing sums of roots of unity: when $3 \mid n$ and n is odd, a projective-collapse argument shows every all-nonzero triad is pairwise disjoint and the coloring constraints decompose; when $2 \mid n$ and $3 \nmid n$, the orthogonality graph on each coordinate plane is a perfect matching, admitting an explicit coloring. The factorization $6 = 2 \times 3$ is consistent with, but logically independent of, the result of Cortez, Morales, and Reyes that $\mathbb{Z}[1/6]$ is the minimal ring supporting KS contextuality in dimension 3.

INTRODUCTION

The Kochen–Specker (KS) theorem [1] establishes that quantum measurements in dimension $d \geq 3$ cannot be explained by noncontextual hidden-variable theories, by exhibiting a finite set of rays admitting no consistent $\{0, 1\}$ -valued coloring. A natural source of KS constructions is the cyclotomic family: for each positive integer n , the alphabet $\{0\} \cup \{\zeta^k : k = 0, \dots, n-1\}$ with $\zeta = e^{2\pi i/n}$ generates a ray set S_n in $\mathbb{C}^3$ via all projectively distinct nonzero vectors with coordinates drawn from this alphabet.

Cabello [2] exhibited a 33-vector, 14-basis KS set in $\mathbb{C}^3$ with coordinates in $\{0\} \cup \mu_6$ (hence lying in S_6), holding the record for fewest bases. Cortez, Morales, and Reyes [4] proved that $\mathbb{Z}[1/6]$ is the minimal ring extension of $\mathbb{Z}$ exhibiting KS contextuality in dimension 3—i.e., the primes 2 and 3 are both necessary.

These results suggest a sharp characterization: KS-uncolorability in the root-of-unity coordinate family should require exactly $6 \mid n$. We prove this. (We emphasize that the relevant structure is the available root subgroup μ_n , not the ambient field: $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\zeta_6)$, yet S_3 is colorable while S_6 is not.)

Theorem 1. *The ray set S_n generated by the n th roots of unity in $\mathbb{C}^3$ is KS-uncolorable if and only if $6 \mid n$.*

PRELIMINARIES

A ray in $\mathbb{C}^3$ is a one-dimensional subspace, represented by a nonzero vector up to scalar multiples. Two rays v, w are *orthogonal* if $\langle v|w \rangle = \sum_k \bar{v}_k w_k = 0$. An orthogonal *triad* is a set of three mutually orthogonal rays (an orthonormal basis up to normalization). A *KS coloring* assigns each ray a value in $\{0, 1\}$ such that (I) orthogonal rays cannot both receive 1, and (II) every triad contains exactly one 1. A ray set is *KS-uncolorable* if no such coloring exists.

For $\zeta = e^{2\pi i/n}$, the ray set S_n consists of all projectively distinct (modulo $\mathbb{C}^\times$) nonzero vectors with coordinates in $\{0\} \cup \{\zeta^j : 0 \leq j < n\}$. We classify rays by *zero pattern*: *axis rays* have two zero coordinates (3 rays); *one-zero rays* have exactly one zero ($3n$ rays: normalizing the first nonzero coordinate to 1 leaves one free phase, and there are 3 choices of zero position); *all-nonzero rays* have no zeros (n^2 rays: normalizing the first coordinate to 1 leaves two free phases).

A key structural observation: in $\mathbb{C}^3$, the inner product $\langle v|w \rangle = \sum_k \bar{v}_k w_k$ has at most three nonzero terms. Each nonzero term is $\zeta^a \zeta^b = \zeta^{b-a}$, an n th root of unity. Therefore only 1-, 2-, and 3-term vanishing sums of n th roots arise as orthogonality conditions in S_n .

PROOF OF THEOREM 1

Sufficiency: an explicit S_6 witness

If $6 \mid n$, then $\zeta^{n/6}$ generates the sixth roots of unity as a subgroup of $\langle \zeta \rangle$, so the alphabet $\{0\} \cup \mu_6$ for S_6 embeds coordinatewise into the alphabet for S_n , and any KS-uncolorable subset of S_6 remains KS-uncolorable inside S_n (any candidate coloring of S_n restricts to a coloring of the subset preserving orthogonality, since orthogonality of fixed rays is intrinsic). It therefore suffices to exhibit a single KS-uncolorable set of rays whose coordinates lie in $\{0\} \cup \mu_6$.

Cabello’s 33-ray witness [2] achieves this. Writing $\omega = \zeta^{n/3}$ for a primitive cube root of unity, the 33 rays in Table I arise as the closure of the Yu–Oh state-independent contextuality set [3] (rays 1–13) under the projective action of the Weyl–Heisenberg generators $X : e_i \mapsto e_{(i+1) \bmod 3}$ and $Z = \text{diag}(1, \omega, \omega^2)$. Cabello proves this set is the unique 33-vector, 14-basis KS set up to unitary equivalence [2]; the rays of Table I are one such realization, with every coordinate equal to 0 or a sixth root of unity (since $\{\pm 1, \pm \omega, \pm \omega^2\} = \mu_6$), so the set embeds in S_6 .

TABLE I. The 33 rays of the Cabello set, with coordinates in $\{0\} \cup \mu_6$ ($\omega = e^{2\pi i/3}$). Rays 1–13 are the Yu-Oh set [3]; rays 14–33 are obtained by closing under the Weyl-Heisenberg generators X, Z .

Yu-Oh 13 rays (real)		
1 (1, 0, 0)	2 (0, 1, 0)	3 (0, 0, 1)
4 (1, 1, 0)	5 (1, -1, 0)	6 (1, 0, 1)
7 (1, 0, -1)	8 (0, 1, 1)	9 (0, 1, -1)
10 (1, 1, 1)	11 (1, 1, -1)	12 (1, -1, 1)
13 (-1, 1, 1)		
Weyl-Heisenberg augmentations		
14 (1, ω , 0)	15 (1, $-\omega$, 0)	16 (1, 0, ω^2)
17 (1, 0, $-\omega^2$)	18 (0, 1, ω)	19 (0, 1, $-\omega$)
20 (1, ω , ω^2)	21 (1, ω , $-\omega^2$)	22 (1, $-\omega$, ω^2)
23 (1, $-\omega$, $-\omega^2$)	24 (1, ω^2 , 0)	25 (1, $-\omega^2$, 0)
26 (1, 0, ω)	27 (1, 0, $-\omega$)	28 (0, 1, ω^2)
29 (0, 1, $-\omega^2$)	30 (1, ω^2 , ω)	31 (1, ω^2 , $-\omega$)
32 (1, $-\omega^2$, ω)	33 (1, $-\omega^2$, $-\omega$)	

The 14 orthogonal bases formed by these rays comprise the axis basis $\{1, 2, 3\}$, nine “mixed” bases each containing one axis ray and two one-zero rays from the corresponding coordinate plane, and four “all-nonzero” bases $B_{11} = \{10, 20, 30\}$, $B_{12} = \{11, 21, 31\}$, $B_{13} = \{12, 22, 32\}$, $B_{14} = \{13, 23, 33\}$, each being a Weyl-Heisenberg orbit of an h -ray (rays 10–13). The full basis listing is in the supplementary material.

Lemma 1 (Cabello [2]). *The 33 rays of Table I admit no KS coloring.*

Proof. We give two independent verifications. *Analytic* (Cabello [2]): Yu and Oh [3] proved that any $\{0, 1\}$ -coloring of rays 1–13 satisfying (I) assigns 1 to at most one “ h -ray” (rays 10–13). The bases $B_{11}, \dots, B_{14}$ are the four Weyl-Heisenberg orbits of those h -rays, so by (II) they contribute four 1’s across the augmented rays. Cabello [2] shows that this assignment is incompatible with the orthogonality edges linking the four h -orbits to the remaining nine bases: four mutually-orthogonal color classes require representatives among only three Yu-Oh slots. *Computational*: a SAT encoding (one variable per ray; basis clauses and pairwise no-two-1’s clauses) is unsatisfiable, verified with Glucose4; DRAT certificate in the supplementary material. $\square$

With the monotonicity argument above, Lemma 1 completes the sufficiency direction. $\square$

Necessity: vanishing-sum analysis

We show that if $6 \nmid n$, then S_n admits a valid KS coloring. Only sums of ≤ 3 terms arise as orthogonality conditions in $\mathbb{C}^3$ (Section 2), so we need only the elemen-

tary 2- and 3-term cases below (longer vanishing sums of roots of unity [5] play no role here).

Lemma 2 (Three-term vanishing sums). *If $u, v, w \in \langle \zeta \rangle$ and $u+v+w=0$, then $3 \mid n$ and $\{u, v, w\} = \{\lambda, \lambda\omega, \lambda\omega^2\}$ for some $\lambda \in \langle \zeta \rangle$, where $\omega = \zeta^{n/3}$. Conversely, $3 \mid n$ suffices.*

Proof. Write $u = \zeta^a$, $v = \zeta^b$, $w = \zeta^c$. If $a = b$ then $u+v+w = \zeta^a(2 + \zeta^{c-a}) = 0$ would force $|\zeta^{c-a}| = 2$, impossible; so the exponents are pairwise distinct. Setting $x = \zeta^{b-a}$, $y = \zeta^{c-a}$ with $x, y \neq 1$, the equation becomes $1 + x + y = 0$. Then $1 = |y|^2 = |1+x|^2 = 2 + 2\operatorname{Re} x$, so $\operatorname{Re} x = -1/2$, forcing $x \in \{\omega, \omega^2\}$ (and y to the other). Hence $\omega \in \langle \zeta \rangle$, requiring $3 \mid n$, and $\{u, v, w\} = \{\lambda, \lambda\omega, \lambda\omega^2\}$ with $\lambda = \zeta^a$. $\square$

Lemma 3 (Two-term vanishing sums). *The equation $\zeta^a + \zeta^b = 0$ has a solution if and only if $2 \mid n$.*

Proof. $\zeta^a + \zeta^b = 0$ iff $\zeta^{b-a} = -1$ iff $-1 \in \langle \zeta \rangle$ iff $2 \mid n$. $\square$

These lemmas, combined with the zero-pattern classification, determine all orthogonalities in S_n . For each pair of rays, the inner product $\langle v|w \rangle$ is a sum of t roots of unity, where t equals the number of coordinates in which both rays are nonzero:

Ray pair	t	Orth. condition*
all-nz vs. all-nz	3	$3 \mid n$
all-nz vs. one-zero	2	$2 \mid n$
one-zero vs. one-zero (same)	2	$2 \mid n$
one-zero vs. one-zero (diff.)	1	never
any vs. axis	0–1	$v_k = 0$

*Necessary divisibility condition for orthogonality between rays of the given type; the exact condition also involves phase congruences (e.g. all-nz vs. all-nz needs the exponent differences to be equilateral mod n).

The “diff.” case has $t = 1$ (the two distinct zero coordinates kill two inner-product terms; the survivor has modulus 1). The axis rays form a triad for all n .

We now treat the three cases where $6 \nmid n$.

Case 1: $3 \nmid n$, $2 \nmid n$

Any triad requires three mutually orthogonal rays spanning $\mathbb{C}^3$. A non-axis triad must contain at least one ray with fewer than two zero coordinates, hence at least one orthogonal pair whose inner product is a 2-term or 3-term sum of roots. Since $3 \nmid n$ forbids 3-term vanishing sums and $2 \nmid n$ forbids 2-term vanishing sums, no such pair can be orthogonal. The sole triad is the axis triad $\{e_1, e_2, e_3\}$, trivially colorable. $\square$

Case 2: $3 \nmid n, 2 \mid n$

By Lemma 2, no all-nonzero orthogonal pairs exist, so no triad can contain two all-nonzero rays.

Lemma 4 (No all-nonzero ray in any triad when $3 \nmid n$). *If $3 \nmid n$, every triad in S_n contains an axis ray.*

Proof. A triad not containing an axis ray consists of rays each having at most one zero coordinate. Since no two all-nonzero rays are orthogonal ($3 \nmid n$), at most one ray is all-nonzero; the other two are one-zero. Two one-zero rays from *different* planes are never orthogonal (Lemma 6), so both one-zero rays r_1, r_2 share the same zero coordinate k and lie in the plane $\{x \in \mathbb{C}^3 : x_k = 0\}$. Since $r_1 \perp r_2$ are nonzero, they are linearly independent (orthogonal nonzero vectors cannot be scalar multiples), hence span the whole plane. Any vector orthogonal to both therefore lies in its orthogonal complement, the k -axis: the third ray v is proportional to e_k , contradicting the assumption that v is all-nonzero. A fortiori, three one-zero rays cannot form a triad (the same plane argument applies to any two, and different planes are excluded by Lemma 6). $\square$

Moreover, axis rays are never orthogonal to all-nonzero rays ($e_k \perp v$ requires $v_k = 0$). Since every triad contains an axis ray, no all-nonzero ray appears in any triad. Every triad has the form $\{v_1, v_2, e_k\}$ where v_1, v_2 are one-zero rays in the k th coordinate plane and e_k is the complementary axis ray.

Lemma 5 (Perfect matching). *For even n , the orthogonality graph on one-zero rays in each coordinate plane is a perfect matching.*

Proof. The one-zero rays in the $z = 0$ plane are projectively $(1, \zeta^p, 0)$ for $p \in \{0, \dots, n-1\}$. Two such rays p, q are orthogonal iff $1 + \zeta^{q-p} = 0$, i.e., $q - p \equiv n/2 \pmod{n}$. Each ray has exactly one partner, giving $n/2$ disjoint edges. $\square$

Lemma 6 (Plane independence). *One-zero rays from different coordinate planes are never orthogonal.*

Proof. If v has $v_i = 0$ and w has $w_j = 0$ with $i \neq j$, then $\langle v|w \rangle = \sum_k \bar{v}_k w_k$ has exactly one nonzero term (at the coordinate $k \neq i, j$), which is a product of roots of unity, hence has modulus 1. For example, $\langle (\zeta^a, \zeta^b, 0) | (\zeta^c, 0, \zeta^d) \rangle = \bar{\zeta}^a \zeta^c + 0 + 0 = \zeta^{c-a} \neq 0$. $\square$

We construct an explicit coloring. Assign $e_1 \mapsto 1$, $e_2, e_3 \mapsto 0$. All one-zero rays in the $x=0$ plane receive 0 (their triads with e_1 already have one 1). In each of the $y=0$ and $z=0$ planes, the matched pairs (Lemma 5) are independently 2-colored: one 1, one 0 per pair. All-nonzero rays receive 0.

Verification of (II): each triad $\{v_1, v_2, e_k\}$ has exactly one 1. For $k = 1$: $e_1 = 1$, both partners 0. For $k = 2, 3$: axis is 0, exactly one of the matched pair is 1.

Verification of (I): we check that no two rays *both assigned 1* are orthogonal. The ray e_1 is orthogonal only to rays with $v_1 = 0$, all of which received 0. Any ray assigned 1 in the $y=0$ plane may be orthogonal to its matched partner in that plane and to one or more all-nonzero rays (via a 2-term cancellation, possible since $2 \mid n$); however the matched partner is assigned 0 by construction, e_2 is assigned 0, and all-nonzero rays are assigned 0. Cross-plane one-zero rays are never orthogonal (Lemma 6). The same holds symmetrically for $z=0$. Constraint (I) is therefore satisfied, and the coloring is valid. $\square$

Case 3: $3 \mid n, 2 \nmid n$

This is the principal case. Let $\omega = e^{2\pi i/3} = \zeta^{n/3}$, a primitive cube root of unity. Three-term cancellations create triads among all-nonzero rays, but we show these triads are pairwise disjoint and the KS constraints decompose. First, since $2 \nmid n$, no 2-term vanishing sums exist, so no all-nonzero ray is orthogonal to any one-zero ray, and no one-zero ray is orthogonal to another one-zero ray. The all-nonzero and zero-bearing sectors are therefore completely decoupled.

Lemma 7 (Projective collapse). *For any all-nonzero ray v in S_n with $3 \mid n$, there are exactly two projectively distinct all-nonzero rays orthogonal to v .*

Proof. Normalize $v = (1, \zeta^p, \zeta^q)$. For all-nonzero $w = (1, \zeta^r, \zeta^s)$,

$$\langle v|w \rangle = 1 + \zeta^{r-p} + \zeta^{s-q}.$$

By Lemma 2, vanishing requires the three terms to form $\{\lambda, \lambda\omega, \lambda\omega^2\}$. Since one term is 1, we need $1 \in \{\lambda, \lambda\omega, \lambda\omega^2\}$, giving $\lambda \in \{1, \omega^2, \omega\}$. All three choices produce the same unordered set $\{1, \omega, \omega^2\}$; what differs is which element is assigned to which of the three terms $1, \zeta^{r-p}, \zeta^{s-q}$. Since the first term is fixed at 1, only two assignments remain: $(\zeta^{r-p}, \zeta^{s-q}) = (\omega, \omega^2)$ or (ω^2, ω) . This gives exactly two projective solutions:

$$\begin{aligned} w_+ &= (1, \omega \zeta^p, \omega^2 \zeta^q), \\ w_- &= (1, \omega^2 \zeta^p, \omega \zeta^q). \end{aligned}$$

(The three λ -values yield only w_+ and w_- up to projective equivalence.) Since $\omega \neq 1$: $w_+ \not\sim v$, $w_- \not\sim v$, and $w_+ \not\sim w_-$. $\square$

Lemma 8 (Triad uniqueness and 3-cycle orbit). *The set $\{v, w_+, w_-\}$ is a triad, and it is the unique triad among all-nonzero rays containing v . Furthermore the map $T_+ : (1, \zeta^p, \zeta^q) \mapsto (1, \omega \zeta^p, \omega^2 \zeta^q)$ satisfies $T_+^3 = \text{id}$*

with $T_+(v) = w_+$ and $T_+^2(v) = w_-$, so the T_+ -orbit of each all-nonzero v has size exactly 3 and equals its unique triad.

Proof. Orthogonality: $\langle w_+ | w_- \rangle = 1 + \omega^2 \cdot \omega + \omega \cdot \omega^2 = 1 + \omega + \omega^2 = 0$ (using $\bar{\omega} = \omega^2$). *Uniqueness:* v has exactly two all-nonzero orthogonal neighbors (Lemma 7); both lie in $\{v, w_+, w_-\}$. *Orbit size:* directly, $T_+^2(v) = (1, \omega^2 \zeta^p, \omega^4 \zeta^q) = (1, \omega^2 \zeta^p, \omega \zeta^q) = w_-$, and $T_+^3(v) = (1, \omega^3 \zeta^p, \omega^6 \zeta^q) = v$. Hence each T_+ -orbit has exactly three elements and coincides with the triad. $\square$

Lemma 9 (One-zero isolation). *For odd n with $3 \mid n$, every one-zero ray has exactly one orthogonal neighbor (the complementary axis ray) and participates in no triad.*

Proof. Let v be a one-zero ray with $v_k = 0$. We show v is orthogonal only to the axis ray e_k . Clearly $\langle v | e_k \rangle = 0$ since both nonzero coordinates of v are multiplied by 0. For the converse, consider any other ray $w \not\sim e_k$:

Case (a): w is all-nonzero. Then $\langle v | w \rangle$ has exactly two nonzero terms (at the positions where v is nonzero), each a root of unity. This is a 2-term sum, which vanishes only if $2 \mid n$ (Lemma 3). Since n is odd, $v \not\sim w$.

Case (b): w is one-zero with $w_j = 0$, $j = k$ (same plane). Then $\langle v | w \rangle$ is again a 2-term sum of roots of unity. Since $2 \nmid n$, $v \not\sim w$.

Case (c): w is one-zero with $w_j = 0$, $j \neq k$ (different plane). Then exactly one coordinate position has both v and w nonzero, so $\langle v | w \rangle$ is a single root of unity, which cannot vanish. Hence $v \not\sim w$ (this is Lemma 6).

Case (d): w is an axis ray e_j with $j \neq k$. Then $\langle v | e_j \rangle = \bar{v}_j \neq 0$ since $v_j \neq 0$ (the only zero coordinate of v is at position k , not j).

Thus $\deg(v) = 1$ in the orthogonality graph, and v cannot participate in any triad (which requires mutual orthogonality of three rays). $\square$

By Lemma 7, every all-nonzero ray v has orthogonal neighborhood exactly $N(v) = \{T_+v, T_+^2v\}$ (two neighbors, no more); applying the same to T_+v and T_+^2v shows their neighborhoods are also contained in $\{v, T_+v, T_+^2v\}$, so this triple is a connected component—a K_3 —with no edges leaving it. The n^2 all-nonzero rays therefore partition into $n^2/3$ isolated triads (the T_+ -orbits), with no orthogonality edges between distinct triads. Since $2 \nmid n$, no all-nonzero ray is orthogonal to any one-zero ray, and no all-nonzero ray is orthogonal to any axis ray ($\langle e_k | w \rangle = \bar{w}_k \neq 0$ when w is all-nonzero). By Lemma 9, one-zero rays participate in no triad; they impose only constraint (I) via their pendant edge to the complementary axis ray. The all-nonzero triads, the axis triad, and the one-zero pendants are therefore mutually decoupled.

We color as follows: in each all-nonzero triad, assign one 1 and two 0's; in the axis triad, $e_1 \mapsto 1$, $e_2, e_3 \mapsto 0$; every one-zero ray gets 0. Axiom (II) holds in each triad by construction; axiom (I) holds because no two

rays assigned 1 are orthogonal (within-triad: by triad structure; cross-triad: no edges; axis-vs-others: e_1 has no all-nonzero neighbors, all one-zero rays received 0). $\square$

Smallest instances.—For $n = 5$ (Case 1), S_5 has 43 rays but only the axis triad $\{e_1, e_2, e_3\}$. For $n = 2$ (Case 2, alphabet $\{0, \pm 1\}$), one-zero rays match into orthogonal pairs (e.g. $(1, 1, 0) \perp (1, -1, 0)$) and the plane coloring applies. For $n = 3$ (Case 3), the nine all-nonzero rays form three disjoint triads— $\{(1, 1, 1), (1, \omega, \omega^2), (1, \omega^2, \omega)\}$ and its two T_+ -images—each colored independently alongside the axis triad.

DISCUSSION

Both primes are necessary for distinct reasons. The identity $1 + \omega + \omega^2 = 0$ ($3 \mid n$) creates orthogonal triads among all-nonzero rays; the element $-1 \in \langle \zeta \rangle$ ($2 \mid n$) links them via 2-term cancellations. Absent either, the orthogonality structure decomposes and an explicit coloring exists. This factorization $6 = 2 \times 3$ is suggestively consistent with—though logically independent of—the result of Cortez, Morales, and Reyes [4] that $\mathbb{Z}[1/6]$ is the minimal subring of $\mathbb{Q}$ realizing KS projection matrices in $\mathbb{C}^3$: theirs concerns realizability of $P_v = vv^\dagger / \|v\|^2$ over subrings, ours colorability over a coordinate alphabet, and neither implies the other. Whether the projective-collapse mechanism of Lemma 7 extends to other algebraic alphabets (Gaussian integers, quadratic integer rings) is open.

Formal verification.—The necessity argument and the sufficiency reduction $S_6 \subseteq S_n$ are machine-checked in Lean 4 [6] against Mathlib v4.30; the KS-uncolorability of the 33-ray witness (Table I) is independently verified by SAT (Glucose4), with certificate in the supplementary material.

Large-language-model tools were used for computational assistance, manuscript preparation, and adversarial proof-checking; the author directed the research and accepts full responsibility for the scientific content. The Lean 4 formalization, SAT instance, and DRAT certificate are available at <https://github.com/michaelkernaghan/contextuality>.

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